

Asher Norland

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EXPERIENCE

Firaxis Games

Graphics Programmer

Civilization VII

- Created core procedural systems for in-game environments, including vegetation placement and building distribution.
 - Implemented parametric curve generation and sampling within vectorized terrain generation.
- Worked in close collaboration with multidisciplinary teams of artists and programmers.
 - Designed and built procedural modifiers to streamline procedural asset hierarchies within an asset editor.
 - Integrated an user-friendly scripting grammar with the engine to support procedural content control.
- Implemented efficient CPU ray tracing algorithms for both individual and batched intersection queries.
- Developed generic and specialized acceleration structures to enable rapid scene updates for ray tracing.
- Managed asset performance via extensive and automated performance testing.
 - Designed and implemented the player-facing graphics and AI benchmarking.
 - Provided quantitative data and analysis to guide engineering and art optimization efforts.

Marvel's Midnight Suns

- Provided launch support for HDR rendering.

Sparks-Glencoe, MD

Nov. 2020 – Oct. 2025

Disbelief

Programmer I

- Provided specialist services to clients in the games and hardware industry within their development workflow.
- Audited Unreal Engine and other software for their applicability to modern consoles.
- Contributed to major AR, VR, and console platforms, including their SDK and subsequent Unreal Engine integrations.
- Optimized, debugged, and created features for the Unreal Engine, including audio, graphical post processing, and the engine's use of the Android platform.

Wolves in the Walls

- Made UE4 engine graphics pipeline changes to support global buffers for stylized rendering
- Addressed performance issues within stylized fragment shaders based on published research

Cambridge, MA

Aug. 2018 – Sept. 2020

RPI – Center for Architecture Science and Ecology

Undergraduate Research Assistant

- Developed web frontend and backend for data visualization funded by the Bill & Melinda Gates Foundation.
- Integrated WebGL support for three-dimensional mesh manipulation to enable architectural visualization.

Troy, NY

Jan. 2017 – Jan. 2018

RPI – Rensselaer Center for Open Source

Team Lead

- Built a simulation and visualization engine with a focus on GPU and CPU parallelism for generic computing tasks.
- Developed a rendering framework for real-time path tracing with dynamic and self-balancing workloads.
- Developed a modified Linear Bounding Volume Hierarchy for ray queries that allows objects of vastly different scales (e.g. galaxies and small satellites) to coexist in a shared, nested coordinate system.
- Implemented heterogeneous volume sampling with variable length ray marching for real-time rendering on the GPU.

Troy, NY

Jan. 2015 – Jan. 2018

EDUCATION

Rensselaer Polytechnic Institute (RPI)

B.S in Computer Science

Troy, NY

Aug. 2014 - May 2018

SKILLS & INTERESTS

Programming Languages: *Proficient* C++, C, Python, CMake, Shading Languages; *Familiar* TypeScript, C#

Technologies: *Proficient* CUDA, Vulkan, DirectX 12, Git, Perforce; *Familiar* MPI, OpenCL

Software: *Proficient* Windows, Nsight, RenderDoc, Unreal Engine, AR/VR platforms; *Familiar* Linux, Switch

Interests: Film, Hiking, Photography, Rock Climbing